Hospital Data Analysis

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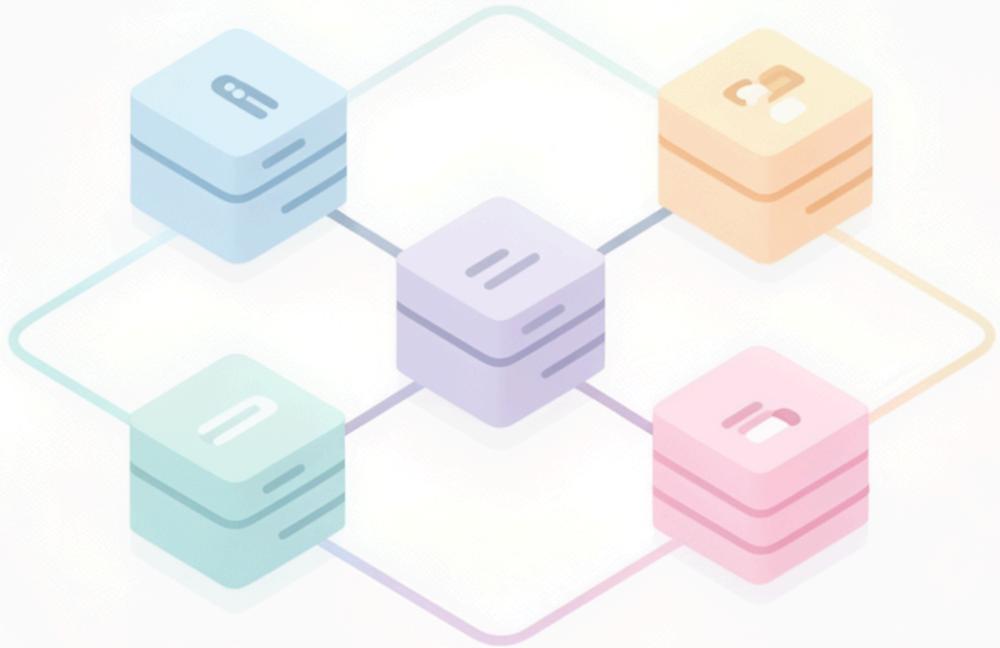
Analyzing Hospital Data for Insight

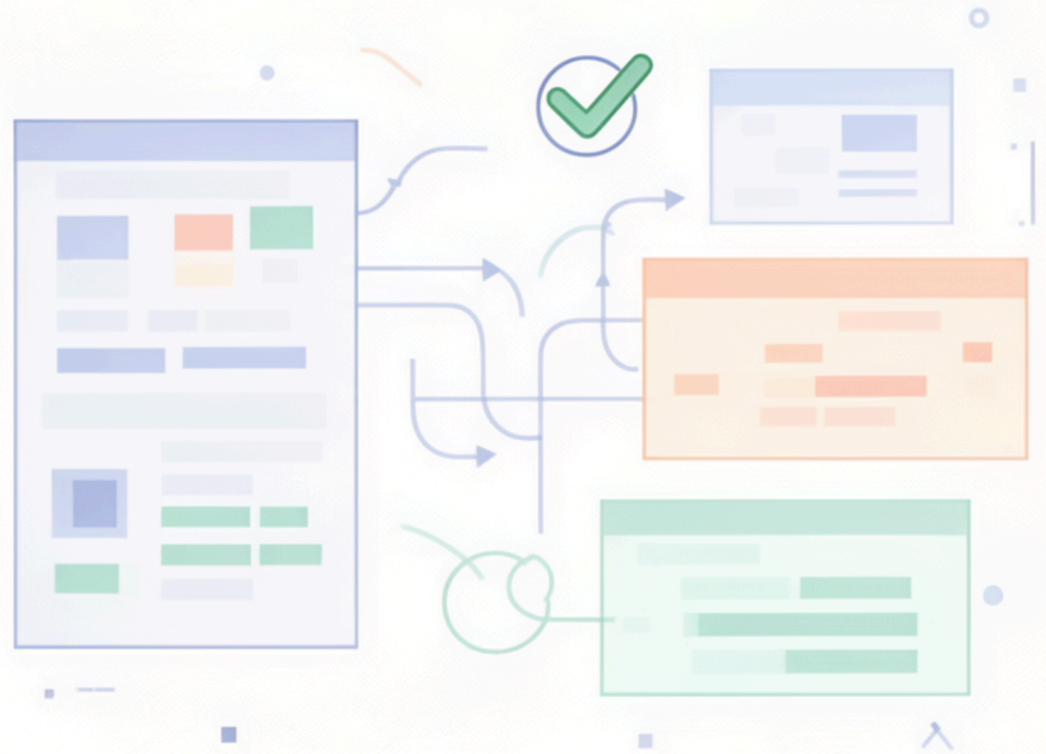
This project focuses on using hospital data analysis to improve decision-making, enhancing operational efficiency and patient care. Key insights drawn from the analysis will support effective management strategies.

Dataset Overview

- Patient: Information on demographics and medical history
- Admission: Records of patient admissions and discharge details
- Billing: Comprehensive billing records for services and treatments

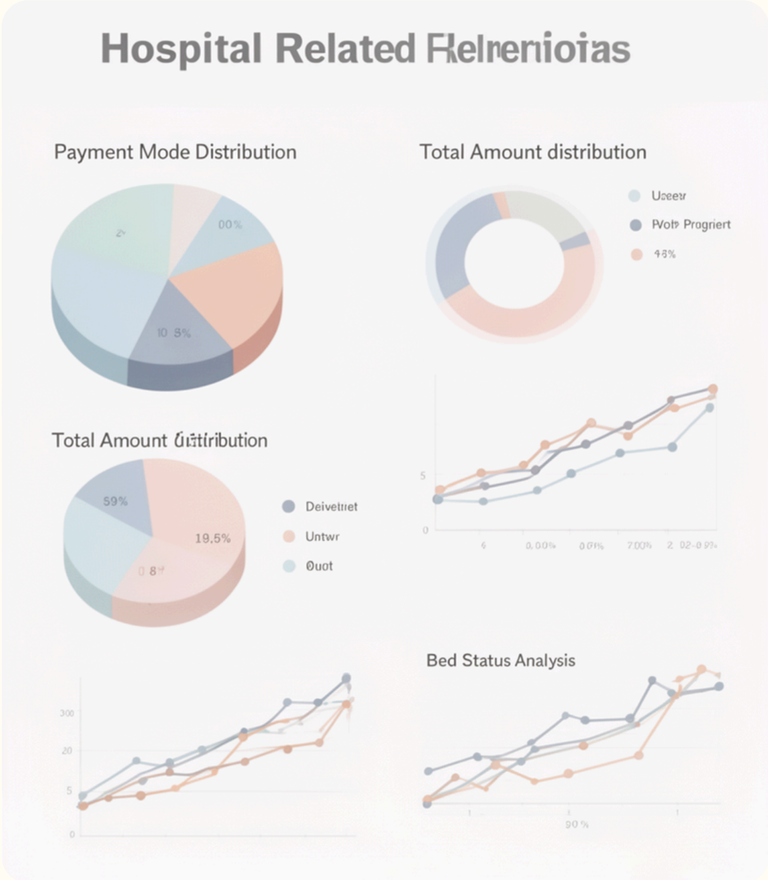


Data Cleaning and Preparation Steps



Effective data analysis begins with thorough cleaning and preprocessing. This involves validating data integrity, removing anomalies, and standardizing formats to ensure accurate and meaningful insights through analysis.

Exploratory Data Analysis Visuals

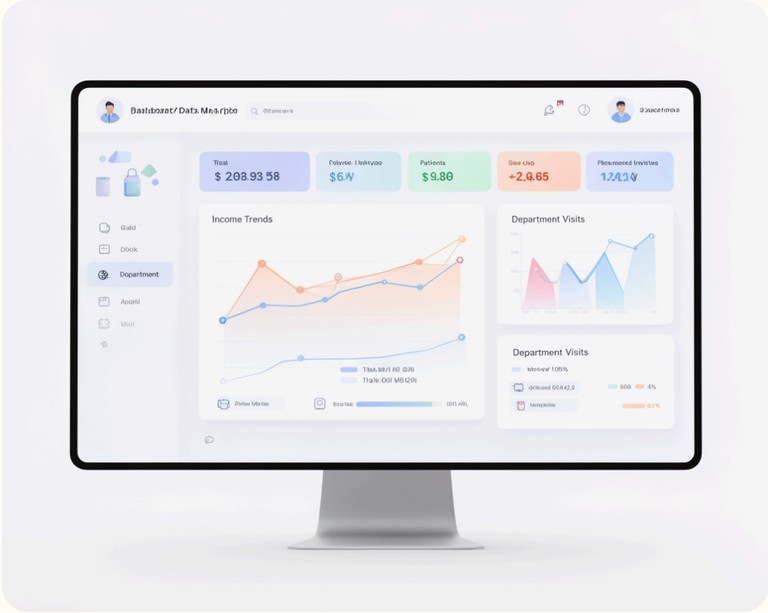


Tools and Workflow Overview

- Python for data cleaning, analysis, and visualization tasks
- SQL for efficient data extraction and aggregating insights
- Power BI for creating engaging and interactive dashboards

Hospital Overview Dashboard Metrics

Key Insights from Dashboard Analysis



Patient Demographics and Behavior

- Distribution of patients by age group and gender
- Analysis of admission types, both emergency and elective
- Most common diseases affecting patient admissions

Department Performance Metrics

- Visits per department indicate overall patient engagement levels
- Average length of stay highlights departmental efficiency and resource usage
- Revenue analysis shows financial contribution from each department

Payment and Revenue Trends

- Total billing amount reflects hospital's overall financial health.
- Average bill amount shows patient care costs per visit.
- Payment status analysis helps manage accounts receivable efficiently.

Medication Prescriptions Analysis

- Most prescribed drugs are identified for patient treatments.
- Drug category usage trends highlight effective therapies and outcomes.
- Average treatment duration helps in resource planning and management.

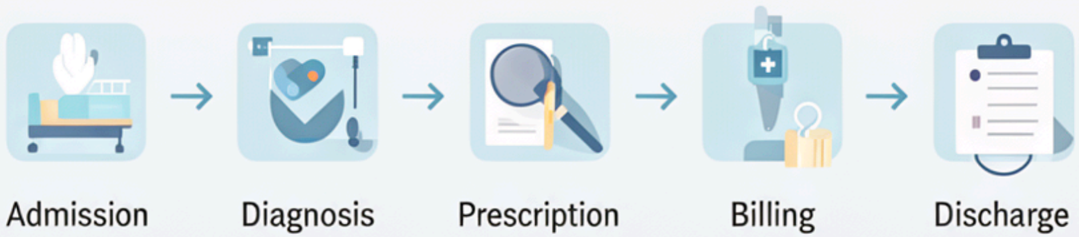
Medication Stock Overview

- Monitoring drug stock levels ensures adequate supply.
- Timely reorder points prevent medication shortages in the hospital.
- Manufacturer distribution impacts procurement and pricing strategies.

Staff Distribution Overview

- Total staff count reflects diverse roles and expertise
- Staff experience contributes to quality patient care outcomes
- Shift distribution ensures optimal departmental coverage and efficiency

Patient Journey Mapping



Understanding the **patient journey** is essential in optimizing healthcare delivery. The process includes admission, diagnosis, prescription, billing, and discharge, facilitating seamless data flow and enhancing patient experience.

Key Insights from the Analysis

Our analysis revealed significant trends in patient demographics, department performance, and revenue drivers, providing actionable insights that can help improve hospital operations and enhance patient care.

Conclusion and Business Impact

This analysis provides actionable insights, enabling hospital management to enhance operational efficiency, improve patient care quality, and optimize resource allocation, ultimately supporting informed decision-making for better health outcomes.

Future Enhancements in Analysis

- **Predictive modeling** to estimate patient length of stay accurately
- **Revenue forecasting** using historical data and trends
- **Readmission prediction** to improve patient management and care

Thank You

For Your Attention and Support